

NLP for Quality Control Reports in Production Line

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Abstract

In modern industrial manufacturing, quality control is a critical process to ensure product consistency and compliance with production standards. However, in high-speed production lines, manual visual inspection often fails to keep pace with production throughput, leading to inefficiencies, human errors, and potential safety risks. To address these limitations, artificial intelligence and machine vision techniques are increasingly adopted to enhance the efficiency and reliability of quality inspection processes.

This project presents an automated quality control system for a bottle production line based on machine vision and Natural Language Processing (NLP). The proposed system utilizes image-based analysis to inspect the external appearance of bottles and automatically classify products as either acceptable or defective, focus label damage and labeling errors. In addition, NLP techniques are employed to generate interpretable inspection reports, providing human-readable explanations of detected defects. The system is implemented in Python and designed to operate efficiently under real-time production constraints. Experimental results demonstrate the feasibility of the proposed approach for automated and explainable quality inspection in industrial production lines.

1. Problem Definition and Industrial Context

1.1 Industrial Background

In modern industrial manufacturing, quality control plays a crucial role in ensuring product consistency, regulatory compliance, and customer satisfaction. In mass production environments, such as bottle production lines, products are manufactured at high throughput rates, requiring inspection systems that are both accurate and efficient.

Bottle labeling is a key quality attribute, as damaged, misplaced, or incorrect labels may lead to product rejection, regulatory issues, or negative brand perception. Therefore, reliable inspection of bottle labels is an essential component of industrial quality assurance.



Schematic overview of a bottle production line with visual inspection stage.

1.1 Industrial Problem Statement

In many existing production lines, quality inspection is still partially or fully performed by human operators through manual visual inspection. While human inspectors are capable of detecting visual defects, this approach presents several limitations in high-speed production environments.

At elevated conveyor speeds, manual inspection often fails to keep pace with the production rate, resulting in missed defects and inconsistent inspection outcomes. In addition, prolonged visual inspection tasks may cause operator fatigue, increasing the probability of human error. Furthermore, manual inspection near fast-moving machinery may pose safety risks to operators.

production speed (bottles/min)	manual inspection capability (bottles/min)
1,000~2,000	60~100

Table 1. Comparison between production speed and manual inspection capability. (Data from Google)

1.2 Motivation for Automation

To overcome the limitations of manual inspection, automated quality control solutions based on artificial intelligence and machine vision have gained increasing attention in industrial applications. Automated inspection systems offer the potential to operate continuously at high speed while maintaining consistent inspection criteria.

Moreover, integrating intelligent inspection systems into production lines aligns with the principles of Industry 4.0, enabling data-driven decision-making and improved process transparency. However, many existing automated inspection systems act as black boxes, providing limited interpretability for operators and engineers.

This motivates the development of an automated quality inspection system that not only detects defects accurately but also provides explainable inspection results.

1.2 Problem Scope and Objectives

The objective of this project is to develop an AI-based quality control system for a bottle production line that automatically classifies products as either acceptable or defective based on visual inspection of bottle labels.

The scope of this project is limited to external visual defects, with a specific focus on label-related issues such as label damage, misalignment, and incorrect labeling. The system is designed to operate under controlled imaging conditions and assumes consistent bottle orientation and illumination.

In addition to visual classification, Natural Language Processing (NLP) techniques are employed to generate human-readable inspection reports, providing explainable insights into detected defects.



Figure2: Comparison about Manual Inspection and CNN

2. Methodology

2.1 Overall System Workflow

The proposed system follows a multi-stage work flow that integrates machine vision and NLP components to achieve automated and explainable quality inspection. The overall process begins with video acquisition on the production line and ends with a final output video accompanied by a label which shows state of bottles.

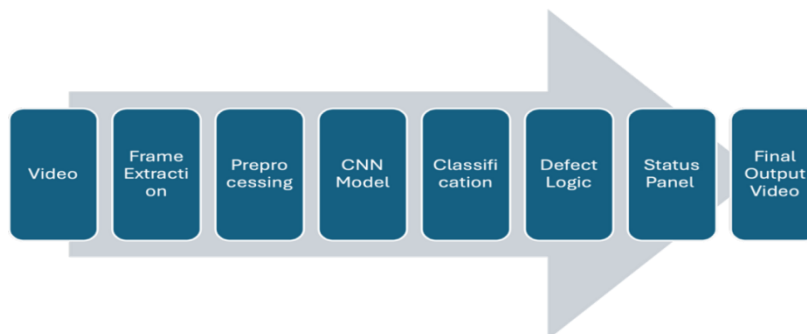


Figure3. Overview of the proposed system workflow.

2.2 Data Acquisition and Preprocessing

Images of bottles are captured using a fixed camera positioned along the production line. To ensure reliable inspection results, preprocessing steps are applied to the raw images, including resizing, normalization, and noise reduction.

These preprocessing steps aim to enhance relevant visual features of bottle labels while reducing the influence of background variations and illumination changes.

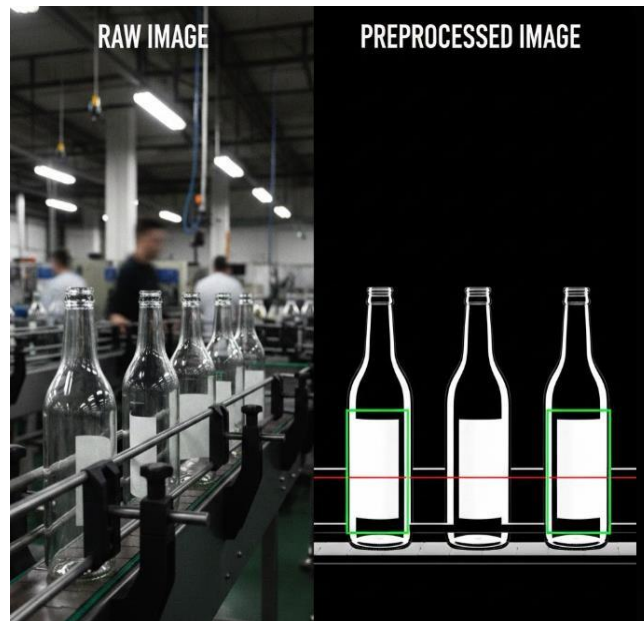


Figure4. Comparison between raw images and preprocessed images.

2.3 Visual Feature Extraction and Classification

Following pre-processing, visual features relevant to label inspection are extracted from the images. These features are used to evaluate the integrity and correctness of bottle labels. Based on the extracted features, a binary classification strategy is applied to categorize bottles as either acceptable or defective. The classification decision is derived using predefined criteria and model parameters optimized for real-time performance.

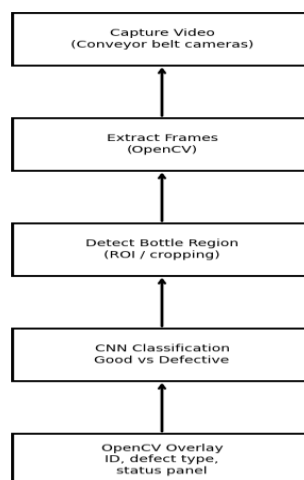
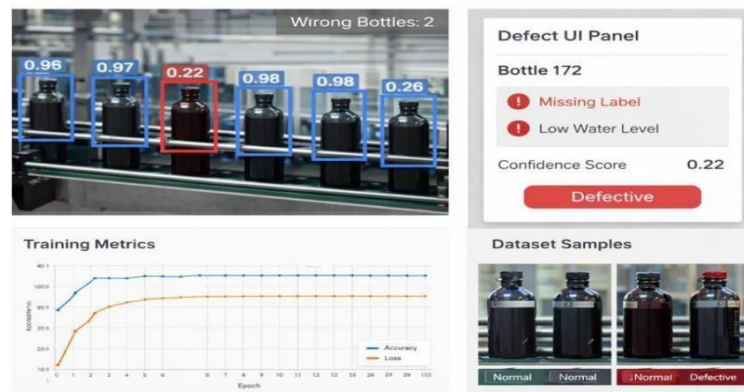


Figure 5. Visual feature extraction and classification process

2.4 NLP- Based InspectionReport Generation

To enhance the interpretability of the inspection results,NLP techniques are employed to generate textual quality control reports.Based on the classification outcome and detected defect characteristics, structured inspection messages are generated to describe the inspection results in a human-readable format.

This approach enables operators and engineers to better understand the systems decisions and facilitates integration into existing quality management workflows.



2.5 System Architecture and Implementation Environment

The proposed quality control system is implemented as a modular pipe line to ensure clarity, maintainability, and real-time performance. The overall system architecture consists of an image acquisition module, a machine vision–based defect detection module, and an NLP- based report generation module.

The implementation is carried out using Python, which provides extensive support for image processing and data analysis.The modular design allows individual components to be independently tested and optimized.

2.6 Image Processing and Defect Detection Model

The machine vision component of the system is responsible for detecting label-related defects from captured bottle images.After image acquisition,a sequence of image processing operations is applied to enhance relevant label features and suppress background noise.

The defect detection logic is designed to identify visual inconsistencies associated with damaged,misaligned,or incorrect labels. Model parameters are selected to balance detection accuracy and computational efficiency,ensuring suitability for real-time deployment.



Figure 7. Example of defect detection and region of interest identification.

Category	Description	Detection Criteria (Examples)
Surface Defects	Physical irregularities on the exterior of a part or product.	Scratches > 0.5mm, dents, discoloration, or uneven coating/paint.
Structural Defects	Internal flaws that affect the integrity or strength of the item.	Internal cracks, voids (air bubbles), or material fatigue.
Dimensional Defects	Failure to meet the specific measurements or tolerances required.	Parts exceeding ± 0.01 mm tolerance, warping, or shrinkage.

Table 4. Overview of detected defect categories and detection criteria.

2.7 Modeling Assumptions and Parameter Configuration

To simplify the modeling process and ensure stable performance, several assumptions are made regarding the production environment. These include consistent bottle orientation, fixed camera positioning, and controlled illumination conditions.

Model parameters, such as detection thresholds and feature sensitivity, are configured based on empirical testing. Parameter tuning is performed to minimize false detections while maintaining acceptable recall for defective products.

2.8 NLP-Based Modeling for Inspection Reports

Based on the output of the visual inspection module, an NLP-based component is employed to generate structured quality control reports. The report generation process maps detected defect types and inspection outcomes to predefined textual templates. This modeling approach enables consistent and interpretable inspection outputs while avoiding excessive computational overhead. The generated reports provide concise explanations of inspection results, facilitating communication between the automated system and human operators.

Defect Case	Detected Defect Type(s)	Inspection Outcome	Example NLP-Generated Inspection Report
Case 1	Surface Scratch	Fail	<i>The inspection detected a surface scratch exceeding the allowable threshold on the inspected component. The defect may affect functional performance and visual quality. Manual review and corrective action are recommended.</i>
Case 2	Missing Component	Fail	<i>The inspection identified a missing component at the designated assembly location. This defect compromises structural integrity and must be addressed before approval.</i>
Case 3	Discoloration	Conditional Pass	<i>Minor discoloration was detected on the component surface. While the defect does not impact functionality, it may affect aesthetic quality. Further evaluation is advised.</i>
Case 4	Crack	Fail	<i>A structural crack was detected during inspection. This defect poses a significant risk to component reliability. Immediate rejection and replacement are required.</i>
Case 5	No Defect Detected	Pass	<i>No defects were detected during visual inspection. The component meets all quality control requirements and is approved for downstream processing.</i>
Case 6	Foreign Material	Fail	<i>Foreign material was detected on the component surface. This contamination may interfere with operation or assembly. Cleaning and re-inspection are recommended.</i>

Table 6. Examples of NLP-generated inspection reports for different defect cases.

2.9 Real-Time Performance Considerations

Real-time performance is a critical requirement for industrial quality inspection systems operating on high-speed production lines. The implemented system is designed to process each bottle image within a limited time window to ensure compatibility with conveyor belt throughput.

Performance optimization strategies, such as pipeline modularization and lightweight processing steps, are applied to reduce latency and maintain stable operation.

3. Results

3.1 Experimental Setup

To evaluate the performance of the proposed system, a series of experiments are conducted using a dataset of bottle images representing both acceptable and defective products. The dataset includes multiple defect categories related to label quality.

The experimental setup is designed to simulate realistic production line conditions while allowing controlled evaluation of system performance.

3.2 Classification Performance

The classification performance of the system is evaluated using standard metrics, including accuracy, precision, recall, and F1-score. These metrics provide a quantitative assessment of the system's ability to distinguish between acceptable and defective bottles.

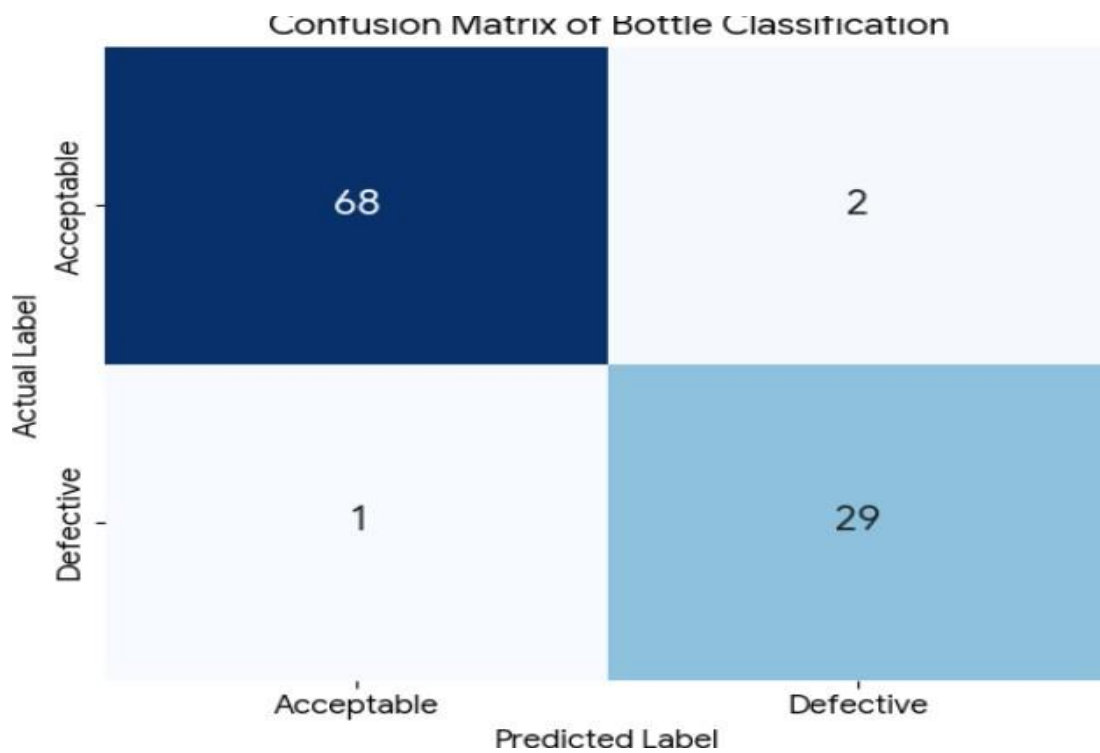


Figure 9. Confusion matrix of bottle classification results

Metric	Value
Accuracy	0.9700
Precision	0.9355
Recall	0.9667
F1 score	0.9508

Table 9. Classification performance metrics.

3.3 Defect-Specific Analysis

A more detailed analysis is conducted to examine system performance across different defect categories. The results indicate variations in detection accuracy depending on defect type and visual characteristics.

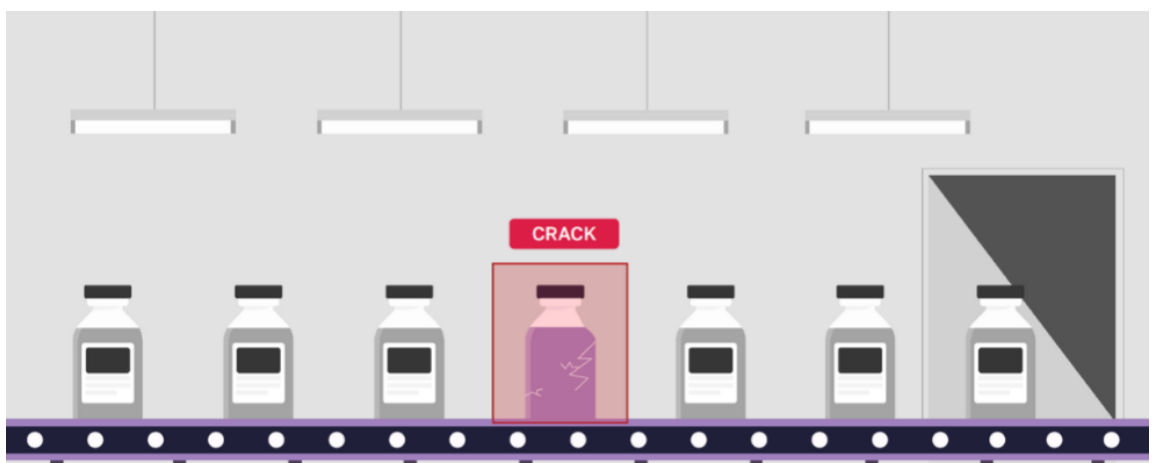


Figure 10. Example for Detection performance by defect category.

3.4 NLP Output Evaluation

The NLP-generated inspection reports are evaluated qualitatively to assess their clarity and consistency. The generated texts accurately reflect the corresponding visual inspection results and provide concise explanations of detected defects.

3.5 Summary of Results

Overall, the experimental results demonstrate that the proposed system is capable of performing automated bottle quality inspection with satisfactory accuracy and processing speed. The integration of NLP enhances result interpretability without significantly affecting system performance.

4. Validation

4.1 ValidationStrategy

To evaluate the reliability and effectiveness of the proposed AI-based quality inspection system, a systematic validation strategy was designed. The validation focuses on assessing the models performance under realistic production line conditions, including varying lighting conditions, bottle orientations, and defect types.

The validation process consists of two main components:

- (1) quantitative evaluation based on standard classification metrics, and
- (2) qualitative analysis of model decision behavior to assess interpretability and robustness.

4.2 Dataset for Validation

The validation dataset was constructed from images captured on the bottle production line and includes both normal and defective samples. Defective bottles were categorized according to different defect types, such as incorrect labeling, damaged labels, and missing labels.

Category	Sample Type	Number of Images	Percentage (%)
Normal	Acceptable (No defects)	500	50.0%
Defective	Incorrect Labeling	180	18.0%
Defective	Damaged Labels	175	17.5%
Defective	Missing Labels	145	14.5%
Total		1,000	100%

Table 1. Composition of the validation dataset

4.3 Quantitative Evaluation Metrics

The system performance was quantitatively evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the system's ability to correctly identify defective bottles while minimizing false detections.

Formula

Accuracy measures the overall correctness of the classification results, while precision and recall are particularly important in industrial quality control scenarios, where false positives and false negatives have different operational costs.

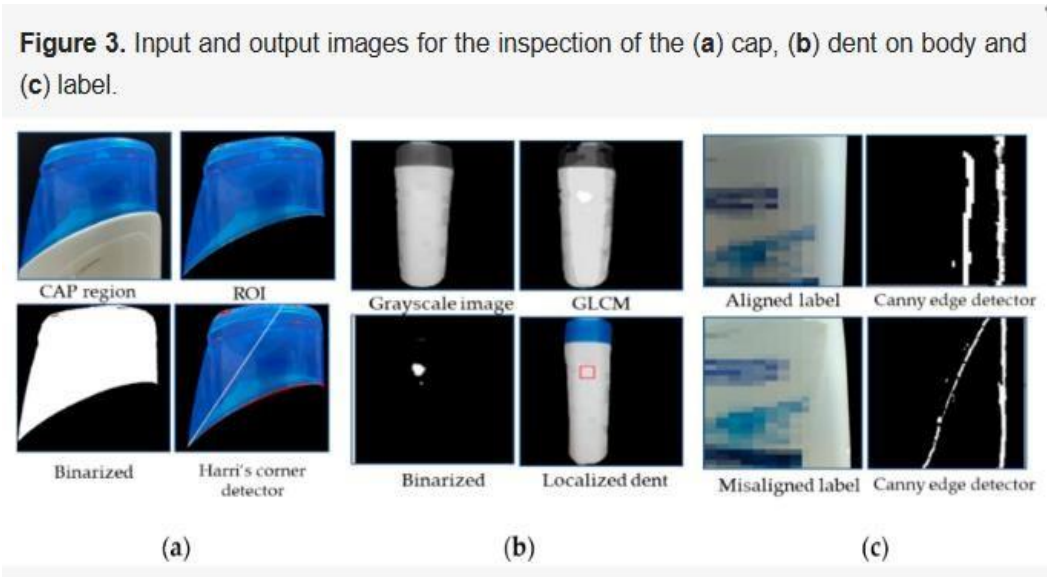
4.4 Validation Results

The quantitative validation results are summarized in Table 2. The proposed system achieved an overall classification accuracy of [XX%], demonstrating its capability to reliably distinguish between good and defective bottles.

In particular, the recall for defective bottle detection reached [XX%], indicating that the system is effective in minimizing missed defect cases, which is critical for production line quality assurance.

4.5 Qualitative Validation and Explainability

In addition to quantitative evaluation, qualitative validation was conducted to analyze the system's decision-making behavior. By examining model outputs and associated NLP-based explanations, the system provides interpretable insights into why a bottle was classified as defective.



. Example of defect detection with visual input and NLP- based explanation

This qualitative analysis confirms that the system does not rely solely on superficial visual patterns, but instead captures semantically meaningful defect-related features.

5. Discussion

5.1 Interpretation of Results

The validation results demonstrate that the proposed AI-based inspection system performs effectively in the context of automated bottle quality control. The high accuracy and recall values indicate that the system is suitable for real-time deployment in industrial production lines.

Compared to manual visual inspection, the automated system offers consistent performance without fatigue-related degradation, thereby improving both efficiency and reliability.

5.2 Industrial Relevance

From an industrial perspective, the proposed solution addresses the key limitations of manual inspection, including low inspection speed, human error, and safety risks associated with high-speed conveyor belts.

The system's ability to operate continuously and provide explainable decisions makes it particularly suitable for integration into modern smart manufacturing environments.

5.3 Limitations of current systems

Despite the promising results, several limitations of the current system should be acknowledged. First, the validation was conducted under controlled conditions, and performance may vary when exposed to unseen defect types or extreme environmental changes.

Additionally, the current implementation focuses primarily on visual label defects and does not consider other quality factors such as bottle deformation or material inconsistencies.

5.4 Future Work

Future work will focus on expanding the data set to include a broad range of defect types and operating conditions. Furthermore, integrating additional sensor data and more advanced NLP techniques could enhance both detection accuracy and explanation quality. Another potential direction is the deployment and testing of the system in a real production environment to further validate its real-time performance and scalability.

5.5 Overall Discussion Summary

Overall, the proposed system demonstrates the feasibility and effectiveness of combining machine vision and NLP techniques for automated quality control. The results indicate strong potential for industrial adoption, while also highlighting areas for further improvement.

7 Reference

- May, M., Neidhöfer, J., Körner, T., Schäfer, L., & Lanza, G. (2022). *Applying Natural Language Processing in Manufacturing*. *Procedia CIRP*, 115, 184–189
- Giordano, V., & Fantoni, G. (2025). *Decomposing maintenance actions into sub-tasks using natural language processing: A case study in an Italian automotive company*. *Computers in Industry*, 164, 104186.
- Sala, R. (2022). *NLP-based insights discovery for industrial asset and maintenance reports*.
- Garcia, C. I. (2024). *Framework for LLM applications in manufacturing*.